

US astronomer Edwin Hubble's meticulous research in the early 20th century redefined our understanding of the cosmos. He proved that the Universe is made up of millions of galaxies, not just the Milky Way (our galaxy), and that it is constantly expanding. These discoveries form the basis of modern cosmology.

Born in Missouri, Edwin Powell Hubble studied mathematics and astronomy at the University of Chicago before relocating to England to read philosophy of law at the University of Oxford to please his father. Returning to the US in 1913, he decided to further his studies in astronomy and, in 1917, received a PhD from Yerkes Observatory. Two years later, he joined the Mount Wilson Observatory in California—home of the Hooker telescope, the world's largest astronomical instrument—as a research astronomer, and spent his entire career there.

In the early 1920s, astronomers were divided into two schools of thought concerning the composition of the Universe: one school believed that the Milky Way was the entire Universe; the other, that it was just one of many galaxies in the Universe. It was Hubble who conclusively proved which theory was correct. Central to his work was the question of whether spiral nebulae (now known as spiral galaxies) lay beyond the Milky Way. In 1923,

“Man **explores the universe** around him and calls the **adventure Science.**”

Edwin Hubble, 1929

MILESTONES

FIRST DISCOVERY

Observes that the Universe is bigger than previously believed while using the Hooker telescope in 1924.

CLASSIFIES GALAXIES

Devises the Hubble Sequence in 1926, which categorizes galaxies by shape, light, and distance.

PUBLISHES KEY PAPER

Links redshift of a galaxy to its distance from Earth in 1929: first evidence that the Universe is expanding.

EINSTEIN VISITS

Meets Einstein at the Mount Wilson Observatory in 1931; shares his evidence of an expanding Universe.

Hubble photographed the nebulae in deep space using the powerful Hooker telescope and was the first to realize they were not gas clouds or distant stars in the Milky Way, as was believed, but galaxies.

